

Supplementary Reproducibility Note

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1 Canonical data inventory

The validated family calculation contains 80 fixed-protocol scan rows: two families, four parameter values per family, and ten paired seeds. The anchor audit contains 100 rows: two family anchors, five graph/control protocols, and ten paired seeds. Every run uses $N = 1000$, $M = 1/2$, $k = 16$, graph-shell radius $r_g = 3$, twelve common radial bins, and the calibrated radius factor 1.15.

All expected family, parameter, seed, and protocol combinations are present. There are zero missing primary or secondary correlations. The maximum radial-angular coordinate mismatch is $1.3322676295501878 \times 10^{-15}$ for the family scans and $8.881784197001252 \times 10^{-16}$ for the anchor audits. Complete vertex sets, connectivity metadata, valid-estimator fractions, mutual-KNN subset checks, and exact or tie-explained edge matching are retained.

2 Zero-parameter consistency

Bardeen $g = 0$ and Hayward $\ell = 0$ reproduce the Schwarzschild/Flamm output row by row for every seed. Coordinate hashes and common-profile hashes agree; the maximum numeric difference is zero at tolerance 10^{-12} .

3 Repository and execution

The repository is github.com/Uxue/PathAnisotropyCurvature. Core estimator functions are under `code/`; resumable headless runners and analysis scripts are under `scripts/`; validation outputs, summaries, and figures are under `results/`; manuscript materials are under `construction_sensitivity_latex_project/` and `submission_jcn/`. Raw numerical results are retained in stable-header CSV files, while completed family/parameter/seed units are stored as resumable Wolfram MX checkpoints in the research repository but excluded from the journal upload ZIP.

Calculations were executed headlessly with Wolfram Language via `wolframscript`. The 100 Bardeen/Hayward checkpoint units required 27,701.3 seconds (7.69 hours), with individual units ranging from 116.5 to 1,099.9 seconds. The preceding Schwarzschild five-protocol audit required 3,636.84 seconds. Manuscript PDFs were compiled with Tectonic; Poppler was used for page rendering and visual quality assurance.